MGMTMSA-400-01/02: Mid-Quarter Exam

Version A

Exam Time: October 28, 2025, 1:00PM – 2:30PM

Disciplines

- Any form of cheating, once identified, will result in a score of **zero** for the exam.
- You may NOT use phones, tablets, computers, ChatGPT, AI glasses, or any other device that has an internet connection. Any use of such device constitutes cheating.

Instructions

- Please answer each question by selecting the correct answer on the supplied bubble sheet. This exam document will NOT be graded.
- On your bubble sheet, fill in the corresponding bubbles for each letter of your name, as well as the bubbles for each digit of your **UID number**.
- Please mark the version of your exam at the top left corner of your bubble sheet.
- This exam has 28 multiple-choice questions, which you must answer within the exam time (90 minutes).
- Some questions ask you to "Select all that apply." For these questions, partial credit is awarded for each correct answer selected. Zero points are awarded if no correct answers are selected, or if any incorrect answers are selected.
- For this exam, you may use (but are not required to use) a scientific calculator. You may also use a paper-based, letter-sized cheat sheet, as long as you do not disrupt others taking the exam.
- While reading the exam, please be careful to consider all possible answers: in some cases, page breaks may cause some of the answers to be displayed on the next page.

Question 1. Suppose there is a random variable $X \sim \mathcal{N}(1,4)$. What is the standard deviation of X?

A. 1

B. 2

C. 3

D. 4

E. None of the above

Answer: B.

Question 2. BrightMart, a large online retailer, recently introduced a personalized recommendation engine to increase customer spending. Historically, the average order value (AOV) per transaction on their platform has been \$50. After implementing the recommendation system, the company collects a random sample of 1,000 transactions from the past month and finds that the new AOV is slightly higher. The team wants to determine whether this represents a meaningful increase in AOV. What is the best statistical analysis they should use?

A. One-sample, one-tailed t-test

B. One-sample, two-tailed t-test

C. Two-sample, one-tailed t-test

D. Two-sample, two-tailed t-test

E. Simple linear regression

Answer: A.

Question 3. A credit card company is designing a new credit card, specifically to appeal to young people who live in cities and who do not currently have credit cards. To that end, they want to know how much this target population typically spends per month, in dollars. To do so, they are planning to run a survey with a market research agency. Specifically, they care about the accuracy of their inference, and want to ensure that the 95% confidence interval is not wider than \$1,000. Based on their existing data from similar customers, they estimate the standard deviation across customers is \$2,000. Given this information, to how many people should they send their survey to ensure they achieve their desired level of precision? *Note*: Some useful percentiles for the standard normal distribution: $z_{0.90} = 1.28$, $z_{0.95} = 1.64$, $z_{0.975} = 1.96$

A. 31

B. 62

C. 124

D. 248

E. 496

Answer: B. The width of the 95% confidence interval of mean monthly spending is

$$2 \times 1.96 \times \frac{2000}{\sqrt{N}}.$$

For it to be smaller than 1000, there must be $N \geq 62$.

Question 4. A startup ran an A/B test to compare two different email subject lines for a promotional campaign. The experimental emails were sent to a test sample of their 200 most active users, with 50% of the sample assigned to each email, at random. After the campaign, they analyzed the results and found that Subject Line A led to a 15% higher click-through rate than Subject Line B, with a p-value of 0.048, analyzed with a two-sample t-test. They concluded that they should send out Subject Line A to the entire customer base. Which of the following is the biggest issue with their study design?

A. The sample is not random, which limits generalizability

B. The p-value is too close to 0.05 to be reliable

C. Lack of a control group makes the results of the test invalid

D. The test sample size is too small

E. They have used the wrong statistical test

Answer: A.

Question 5. Which of the following analyses is supervised learning?

A. Walmart groups customers into segments based on their purchase patterns

B. Outlook classifies whether a new email is spam or not spam based on the email's text

C. Wayfair identifies the key features of its furniture products using image pixels

D. Spotify discovers groups of songs with similar acoustic features

E. All of the above are examples of unsupervised learning

Answer: B. Unsupervised learning does not have an output variable for prediction.

Question 6. Suppose there are two random variables, X and Y, such that $X \sim \mathcal{N}(1,3)$ and $Y \sim \mathcal{N}(1,1)$. The correlation coefficient between X and Y is 0.6. What is the expected value (population mean) of the random variable $X - 2Y$?

A. 1

B. 2.6

C. 3

D. There is not enough information

E. None of the above

Answer: E. The mean should be $1 - 2 \times 1 = -1$.

Question 7. Continue with the same random variables as in the last question: $X \sim \mathcal{N}(1,3)$, $Y \sim \mathcal{N}(1,1)$, and $\text{corr}(X,Y) = 0.6$. What is the variance of the random variable $X - 2Y$?

A. 5

B. 7.16

C. 7.6

D. There is not enough information

E. None of the above

Answer: E. The covariance between X and Y is $\sqrt{3} \times \sqrt{(1)} \times \text{corr}(X,Y) = 1.04$. The variance of $X - 2Y$ should be $\text{Var}(X) + 4\text{Var}(Y) - 4\text{Cov}(X,Y) = 2.84$.

Question 8. Suppose you are interested in estimating the average household income in Los Angeles County. You collect random samples of household incomes from across the county. Which of the following statements are true? **Select all that apply.**

A. Compared to the normal distribution, the distribution of household income in the population is likely to be have a heavy tail on the right (for higher income)

B. The sample mean of household incomes will be approximately normal when the sample size is large

C. The Central Limit Theorem guarantees that the individual incomes in the sample will be normally distributed when the sample size is large

D. As the sample size increases, the sample mean will tend to get closer to the true median of the household income in Los Angeles county

E. None of the statements above are true

Answer: AB.

Jane's All You Can Eat Restaurant

Jane runs an all-you-can-eat restaurant that charges $19 per customer. From her thousands of records over the past year, she finds that her **cost per customer** (including food and labor) follows a normal distribution with a mean of $17.50 and a standard deviation of $3.00.

We'll help Jane make a few predictions in Questions 9-11. Assume that there is no inflation in the past year.

Note: Some useful percentiles for the standard normal distribution: $z_{0.90} = 1.28$, $z_{0.95} = 1.64$, $z_{0.975} = 1.96$

Question 9. What is the best point predictor of the cost for the next customer?

A. 3.00

B. 17.50

C. 20.50

D. 22.50

E. 23.38

Answer: B.

Question 10. Jane wants to find a 90% prediction interval of customer costs, centered around the mean ($17.50). In other words, 90% of customers' costs should fall within this range. What is the width of that interval, i.e., the distance between the upper and lower bounds?

A. 3.84

B. 4.92

C. 7.68

D. 9.84

E. 11.76

Answer: D. The width is

$$2 \times z_{0.95} \times \text{sd}(Y) = 9.84$$

Question 11. Suppose customers are i.i.d. On one day, Jane serves 100 customers, and on another, she serves 250 customers. Which day is more likely to have a higher average

profit **per customer** (price minus cost per customer)?

A. The day with 100 customers

B. The day with 250 customers

C. Both days are equally likely

D. Not enough information

E. None of the above

Answer: C. On both days, the expectation of the average profit is $1.5.

Evaluating Advantage+

Advantage+ is Meta's AI-powered ads product that automates and optimizes campaign setup and delivery. Beginning on 07/01/2024, a U.S. advertiser adopted Advantage+ for all campaigns.

This advertiser would like to understand how much Advantage+ affected daily sales using linear regression with the least squares criterion. The regression is conducted usingdaily data from 01/01/2024 to 12/31/2024. The output variable is `log_sales` (log of dollars per day). The input variable `advantage` is is a binary variable equal to 1 if this advertiser uses Advantage+ in the corresponding day, and 0 otherwise.

The `R` output from the least squares regression is shown below. Some entries are intentionally omitted—you may need to infer them. Questions 12–15 refer to this setup and output.

```
> summary(lm(log(sales) ~ advantage, data = sales))

Call:
lm(formula = log_sales ~ advantage, data = sales)

Residuals:
     Min       1Q   Median       3Q      Max
-2.20022 -0.54692 -0.00463  0.45543  2.43485

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  (--a--)    0.04101 (--b--)   <2e-16
advantage    0.08612    (--c--)   2.091  (--d--)

Residual standard error: 0.7845 on 364 degrees of freedom
Multiple R-squared: 0.01186
F-statistic: 4.371 on 1 and 364 DF,  p-value: 0.03726
```

```
> cat("The mean of advantage is", mean(sales$advantage),
        "and the sample standard deviation is", sd(sales$advantage))
The mean of advantage is -0.0071 and the sample standard deviation is 0.9967

> cat("The mean of log sales is", mean(sales$log_sales),
       "and the sample standard deviation is", sd(sales$log_sales))
The mean of log sales is 7.8415 and the sample standard deviation is 0.7881

> cat("The mean of absolute sales is", mean(sales$sales),
         "and the sample standard deviation is", sd(sales$sales))
The mean of absolute sales is 3511.7 and the sample standard deviation is 3480.0
```

Question 12. What is the estimated value of the intercept (in two decimal places), i.e., the value of `(--a--)`?

A. 0.93

B. 1.86

C. 7.84

D. 15.69

E. 37.50

Answer: C. Use the fact that fitted line under the least squares criterion should cross the sample mean.

Question 13. What is the estimated standard error of the slope on `advantage` (in two decimal places), i.e., the value of `(--c--)`?

A. 0.04

B. 0.05

C. 0.12

D. 0.20

E. 0.23

Answer: A. You can use standard error of regression and sample standard deviation for calculation. The most straightforward way, however, is to use the t-statistic: it is the ratio of estimated coefficient and the standard error. Therefore, the standard error is 0.04.

Question 14. The relatively small value of R^2 implies which of the following are true? **Select all that apply**.

A. The model is not valid

B. Variable `advantage` explains only a small share of the variation in `log_sales`

C. The advertiser should not use Advantage+

D. The correlation between `advantage` and `log_sales` is not strong

E. We should switch the dependent variable to unlogged `sales`

Answer: BD. R-squared is the square of correlation coefficient between X and Y in simple linear regressions.

Question 15. Which of the following would pose the most serious threat to establish causality using the estimation results?

A. In 2022 and 2023, the advertiser used two different agencies to manage display ads

B. Sales often spike during Thanksgiving and Christmas

C. The advertiser's customers represent a niche market

D. The advertiser randomly runs promotions across 2024

E. The advertiser plans to launch a new product next year

Answer: B.

Survey Conduct Design

Suppose you are a pricing manager at Pepsi. You would like to determine whether to adjust the price of the 20-ounce bottle of Diet Pepsi in Sioux City, Iowa, a city with about 90,000 residents. To inform this decision, you plan to conduct a survey to estimate consumers' willingness to pay (WTP) for this product.

In Questions 16–17, you and your analysts aim to align on the appropriate design and implementation of this survey.

Question 16. Before the survey begins, your analyst proposes several ideas about how sample size and distributional assumptions may affect inference. **Select all statements that are correct.**

A. Because Sioux City is relatively small and its residents may have similar preferences, a smaller sample may be sufficient to obtain an estimate of the population's average willingness-to-pay that is as accurate as in another city (e.g., New York City) with more diverse preferences

B. A larger sample size generally allows for more precise inference about the average WTP

of residents

C. Regardless of the sample size, we cannot make valid inferences about the population average WTP if the underlying distribution of WTP is not normal

D. Regardless of sample size, we cannot conduct hypothesis tests about the population mean unless WTP follows a normal or t distribution exactly

E. None of the above

Answer: AB.

Question 17. You plan to collect survey data such that all participants are independent and identically distributed (i.i.d.). Which of the following survey practices **does not violate** the i.i.d. assumption?

A. Skipping participants who only drink Coke

B. After surveying 10 white residents in a row, targeting an African American respondent so that the sample's demographics match the city's population

C. After finding a participant with a high WTP, looking for another participant with a low WTP to reduce sampling error

D. Including respondents from neighboring towns outside Sioux City in the survey

E. All the practices above violate i.i.d.

Answer: E.

Test the Effectiveness of GLP-1

GLP-1 (Glucagon-Like Peptide-1) are medications to regulate appetite and increase insulin production. They are commonly used for weight loss. Jonathan is considering starting a GLP-1 medication for weight loss. Considering the relatively high price of the treatment, he will only start if the average 6-month effectiveness is at least 15% body-weight reduction. You, as Jonathan's doctor, collect a small pilot sample of 10 patients who recently completed 6 months on GLP-1. Their body-weight losses (measured in percentage points) are as below:

$$15.2,\ 14.1,\ 15.7,\ 15.4,\ 15.0,\ 17.2,\ 16.1,\ 14.9,\ 16.6,\ 14.8$$

Questions 18–20 ask you to determine whether you would recommend the prescription for Jonathan.

Note: Some useful percentiles for t-distributions: $t^9_{0.90} = 1.38, t^9_{0.95} = 1.83, t^9_{0.975} = 2.26, t^{10}_{0.90} = 1.37, t^{10}_{0.95} = 1.81, t^{10}_{0.975} = 2.23$

Question 18. What is the mean body-weight loss (in percentage points) from the pilot sample?

A. 15.0

B. 15.4

C. 15.5

D. 16.0

E. None of the above

Answer: C.

Question 19. Which of the following is the best null hypothesis for determining whether to prescribe GLP-1 to Jonathan?

A. The average 6-month body-weight reduction for GLP-1 users is equal to 15%

B. The average 6-month body-weight reduction for GLP-1 users is lower than 15%

C. Jonathan's predicted 6-month body-weight reduction from GLP-1 is equal to 15%

D. Jonathan's predicted 6-month body-weight reduction from GLP-1 is lower than 15%

E. Jonathan's predicted 6-month body-weight reduction from GLP-1 is greater than 15%

Answer: B.

Question 20. You conduct a t-test for Jonathan, and find that the t-statistic is 1.71. What is your prescription decision for Jonathan at a 95% confidence level?

A. There is sufficient statistical (or data) evidence to support prescribing GLP-1 to Jonathan

B. There is sufficient statistical (or data) evidence to reject prescribing GLP-1 to Jonathan

C. There is not enough statistical (or data) evidence to support prescribing GLP-1 to Jonathan

D. We need more data for this decision

E. None of the above

Answer: C.

Election Research Using Regressions

Suppose you are a political science researcher studying how demographic factors shape Americans' political preferences. You are particularly interested in how the following

county-level variables relate to support for Donald Trump in the 2024 U.S. presidential election: (1) average household income, (2) share of residents with a bachelor's degree, and (3) how rural the county is (measured by a *rural index* between 0 and 1).

Your research assistant collects the 2024 election results and runs three separate least-squares regressions, each using Trump's county-level vote share as the dependent variable. This variable varies between 0 and 1. Each regression includes one of the three explanatory variables listed above as its only input. The results are summarized below.

	(1)		(2)		(3)	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
Log(Avg. household income in dollars)	-0.14	0.03				
Share with bachelor's degree			-1.44	0.15		
Rural index					0.83	0.35
(Intercept)	2.12	0.22	1.08	0.09	0.14	0.05
R-squared R^2	0.44		0.37		0.60	
Number of Observations	3145		3145		3145	

In the regression table, `Coef` stands for estimated parameters, and `S.E.` refers to standard errors.

In Questions 21-24, you will answer some questions related to these regression results. Throughout these questions, **regression (1)**, **regression (2)**, and **regression (3)** refer respectively to models that use: (1) log(average household income), (2) share of residents with a bachelor's degree, and (3) the rural index as the input variable. The output variable is always the vote share of Donald Trump.

Note: Some useful percentiles for the standard normal distribution: $z_{0.90} = 1.28$, $z_{0.95} = 1.64$, $z_{0.975} = 1.96$

Question 21. Which variable appears to have the strongest correlation with Trump's county-level vote share?

A. Log(average household income)

B. Share of residents with a bachelor's degree

C. Rural index

D. The intercept

E. There is not enough information

Answer: C. Regression (3) has the highest R^2, and R^2 is the square of the correlation coefficient. Notice that the slope couldn't be compared across different input variables—they are not unitless!

Question 22. What is the 95% confidence interval for the slope coefficient in regression (3), corresponding to the *rural index*? Round your answer to two decimal places.

A. $[0.14, 1.52]$

B. $[0.23, 1.43]$

C. $[0.25, 1.41]$

D. $[0.82, 0.84]$

E. None of the above

Answer: A. The upper bound of the 95% confidence interval is $0.83 + 0.35 \times z_{0.975} = 1.52$. We can do similar calculation for the lower bound.

Question 23. In Philadelphia County, Pennsylvania, Trump received 19.94% of the total vote, and the county's share of residents with a bachelor's degree was 34.6%. Using regression (2), where the share with a bachelor's degree is the input variable, compute the residual for this data point from Philadelphia and round it to two decimal places.

A. -0.38

B. -0.30

C. 0.38

D. 0.70

E. 0.78

Answer: A.

Question 24. Suppose your assistant redefines the income variable in regression (1) so that it measures *log(average household income in thousands of dollars)* instead of in dollars. For example, a county with an average household income of \$80,000 would now have $\log(80)$ instead of $\log(80000)$ as its value for the input variable into regression (1). Compute the new estimated slope for this regression, rounding to two decimal places.

A. -7.05

B. -0.14

C. -0.02

D. -0.00

E. There is not enough information

Answer: B.

Used Car Pricing

Suppose you are providing consulting service to Marissa, a used-car dealer. Marissa asks you to help price vehicles. In turn, she gives you a dataset with 475 observations. The dataset contains price (in thousand dollars), mileage (in thousand miles), and a luxury index (a continuous score varying from 0 to 1). From the sample, you learn that the correlation between price and mileage is -0.65, and the correlation between price and luxury index is 0.45. The sample standard deviation of mileage is 1.95, and that of the luxury index is 0.15.

In Questions 25–28, use these facts and what you know about simple linear regression.

Note: Some useful percentiles for the standard normal distribution: $z_{0.90} = 1.28$, $z_{0.95} = 1.64$, $z_{0.975} = 1.96$

Question 25. Which of the following statements are true? **Select all that apply.**

A. The linear correlation between price and mileage is stronger than that between price and luxury index

B. The correlation between price and mileage is significantly lower than 0

C. Mileage explains more variation in prices than the luxury index does

D. All of the above statements are wrong

Answer: ABC.

Question 26. You run the following two linear regressions with the least squares criterion, where the output variable is car price:

- Regression (a): mileage as an input variable
- Regression (b): luxury index as an input variable

Which estimation will exhibit a larger (in absolute magnitude, or distance from zero) estimated slope?

A. (a)

B. (b)

C. They must be equal in magnitude

D. There is not enough information

E. None of the above

Answer: B.

Question 27. As in the last question, you run two different least-squares linear regressions, with (a) using mileage as an input variable, and (b) using luxury index as an input variable. Which of the estimated slope would have a larger standard error?

A. (a)

B. (b)

C. They must be equal in magnitude

D. There is not enough information

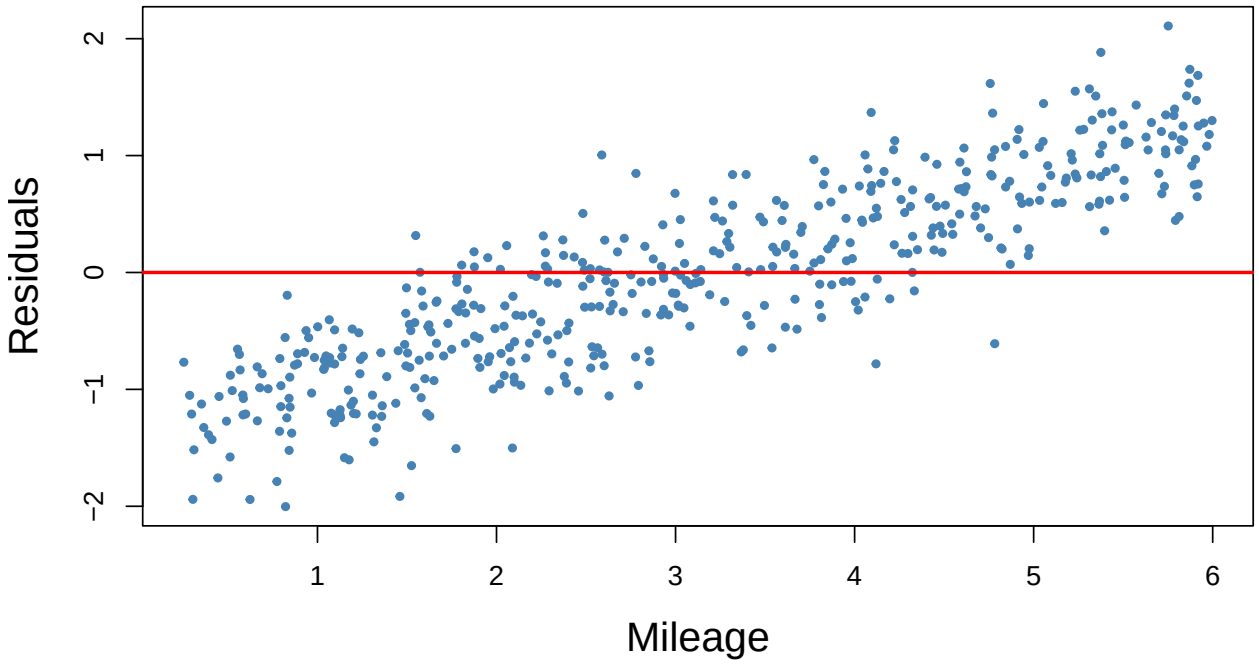
E. None of the above

Answer: B.

Question 28. You firstly use mileage as the input variable. You firstly assume that

$$\hat{y}_i = 5.00 - 0.75 \times x_i,$$

where y is price and x is mileage. You compute the residuals using this line and plot the residuals against x as below:



The mean of the residuals plotted above is zero. Now, suppose you would like to refit the line using least squares criterion. Denote the newly fitted line as $\hat{y} = b_0 + b_1 \times x$. What is the relationship between b_1 and -0.75?

A. $b_1 > -0.75$

B. $b_1 = -0.75$

C. $b_1 < -0.75$

D. There is not enough information

E. None of the above

Answer: A.